The set of matchings avoiding a short-chord-free pattern is closed under Knuth equivalence

Avichai Marmor

doi:10.5802/alco.351; CC BY 4.0

Editorial note. The counted statement is the author's labelled lemma environment at publisher-source.tex lines 1300-1303; its proof environment opens at line 1304, immediately after \end{lemma}, so the pairing is by adjacency and the retained proof range 1304-1307 is the whole environment.

Editorial note. The definition def:pattern avoidance (source lines 1262-1279) is retained as context because the counted proof uses the notion of pattern containment introduced there; it is the only outside label of the retained proof.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label lem:avoiding set of short chord free matching is knuth like closed.

Editorial scope. One original result of the pinned publisher article, counted once: closure under Knuth equivalence of the set of matchings avoiding a short-chord-free pattern (author label lem:avoiding set of short chord free matching is knuth like closed).

Allowed standard prerequisites (editorial).

- Pattern containment and avoidance. A matching m_2 contains m_1 when an order-preserving choice of vertices carries the chords and unmatched vertices of m_1 to those of m_2, and a set of matchings avoids m_1 when none of its members contains it.. It is the notion appearing in the statement.
- Elementary Knuth transformation. Knuth equivalence is generated by a local rewriting of three consecutive chords of a matching.. The proof only has to check one such transformation.
- Short-chord-free pattern. The pattern m is assumed to have no short chord, so no chord of m joins two consecutive vertices.. It is what makes the case distinction in the proof exhaustive.

Definition 0.1. Let $m_1 \in \mathcal{M}_{N_1}$ and $m_2 \in \mathcal{M}_{N_2}$ for some positive integers $N_1 \leq N_2$. We say that m_2 contains the pattern m_1, if there exist indices $1 \leq i_1 < \cdots < i_{N_1} \leq N_2$ such that the following holds:

- For all $1 \leq j < j' \leq N_1$,
$$(j, j') \in m_1 \iff (i_j, i_{j'}) \in m_2.$$
- For all $1 \leq j \leq N_1$,
$$(j) \in m_1 \iff (i_j) \in m_2.$$

Otherwise, we say that m_2 avoids m_1.

For two sets of matchings $\mathcal{M}_1 \subseteq \mathcal{M}_{N_1}$ and $\mathcal{M}_2 \subseteq \mathcal{M}_{N_2}$, denote by $\mathcal{M}_2(\mathcal{M}_1)$ the set of matchings in $\mathcal{M}_2$ that avoid every matching in $\mathcal{M}_1$. In addition, denote $\mathcal{M}_2(m) := \mathcal{M}_2(\{m\})$ for a matching m.

Lemma 0.2. Let $m \in \mathcal{M}_N$ be a short-chord-free matching, and let N', f' be nonnegative integers. Then the pattern-avoiding set $\mathcal{M}_{N',f'}(m)$ is closed under Knuth equivalence.

Proof. Let $m'_1 \in \mathcal{M}_{N',f'}$ be a matching, and let $m'_2 \in \mathcal{M}_{N',f'}$ be the result of applying an elementary Knuth transformation on m'_1. Assume that m'_1 contains the pattern m. It suffices to prove that m_2 contains m too. By Definition 0.1, there exist indices $1 \leq i_1 < \cdots < i_N \leq N'$ such that $(j, j') \in m \iff (i_j, i_{j'}) \in m'_1$ for all $1 \leq j < j' \leq N$ and $(j) \in m \iff (i_j) \in m'_1$ for all $1 \leq j \leq N$. Without loss of generality, we may assume that $(i, i+1) \in m'_1$ and m'_2 is obtained from m'_1 by interchanging the chord $(i, i+1)$ with the vertex $i+2$. The pattern m is short-chord-free, so $i, i+1 \notin \{i_1, \ldots, i_N\}$. Therefore, by considering the index set $\{i_1, \ldots, i_N\}$ if $i+2 \notin \{i_1, \ldots, i_N\}$ and the set $\{i\} \cup \{i_1, \ldots, i_N\} \setminus \{i+2\}$ otherwise, we obtain that the matching m'_2 contains m as well. □